

5(a)	<i>any four from:</i> 1 find, nucleotide / base / gene / DNA, sequence ; 2 from database / using bioinformatics / from gene library ; 3 make the gene (chemically) using nucleotides ; 4 extract mRNA from cell, making the protein / expressing the gene ; 5 use reverse transcriptase (with mRNA) to make cDNA ; 6 use DNA polymerase (with cDNA) to make dsDNA ; 7 cut out / extract / isolate, the gene from DNA ; 8 using restriction enzyme ;	4
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Question	Answer	Marks
5(b)	any three from: ora for animal insulin 1 large scale supply / supply can match demand ; 2 no / less, likely to cause, immune / allergic, response or fewer side-effects ; 3 faster to act / smaller dose needed ; 4 no / less, likely for (insulin) tolerance to occur ; 5 no / less, risk of infection / disease ; 6 no / less, ethical / religious objections or suitable for, vegetarians / vegans ; 7 AVP ; e.g. cheaper to produce / easier to purify	3
5(c)	DNA from two different sources (joined together) or DNA from (human) gene joined to DNA from, bacteria / plasmid ;	1

Question	Answer	Marks
5(d)	any four from: 1 find, nucleotide / base / gene / DNA, sequence (of human insulin gene) ; 2 from database / using bioinformatics / from gene library ; 3 gene editing ; 4 insertion / deletion / replacement, of <u>nucleotide(s)</u> ; 5. at specific sites (of the gene) ; 6 (results in) changes to the, (DNA) triplet / codon ; 7 (so codes for a) different amino acid ; 8 AVP ; e.g. detail of gene editing / CRISPR / Cas9	4